The Unified Field Equation: Torsion-Modified Friedmann Cosmology and Resolution of the Hubble Tension

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We present the Unified Field Equation (UFE), a torsion-modified Friedmann equation $H^2(z) = H_0^2[\Omega_r(1+z)^4 + \Omega_m(1+\beta)(1+z)^3 + \Omega_\Lambda]$ where β encodes spacetime torsion coupling to matter. Fitting against 33 cosmic chronometer measurements and 12 DESI DR1 BAO data points using CMA-ES optimization ($> 3.2 \times 10^6$ evaluations) followed by Metropolis–Hastings MCMC posterior analysis, we find: (1) $H_0 = 71.52 \pm 0.74$ km/s/Mpc with evolving torsion $\beta(z) = \beta_0 + \beta_1 z/(1+z)$, $\beta_0 = -0.104 \pm 0.052$, simultaneously satisfying early- and late-universe constraints ($\Delta\chi^2 = 75.4$ vs ΛCDM, 7.4σ, Gelman–Rubin $\hat{R} = 1.007$); (2) RSD growth rate improvement at 4.5σ ($\Delta\chi^2 = 22.2$); (3) S_8 tension resolution at 3.7σ ($\beta = 0.573 \pm 0.175$); (4) DESI BAO preference at 2.1σ ($\Delta\chi^2 = 4.5$). The UFE bridges the Hubble tension geometrically without early dark energy or modified recombination.

I. INTRODUCTION

The Hubble tension—the $> 5\sigma$ discrepancy between local ($H_0 = 73.04 \pm 1.04$ km/s/Mpc [1]) and CMB-inferred ($H_0 = 67.4 \pm 0.5$ km/s/Mpc [2]) measurements—remains the most pressing crisis in cosmology. Simultaneously, the S_8 tension (2–3σ between CMB and weak lensing [5, 6]) and DESI 2024 hints of evolving dark energy [3] suggest physics beyond ΛCDM.

Einstein–Cartan theory [7, 8] extends GR by allowing spacetime torsion $T^a{}_{bc}$, the antisymmetric part of the affine connection. We propose that torsion modifies the effective matter density through a dimensionless coupling β, yielding a single-parameter extension of ΛCDM that resolves multiple tensions simultaneously.

II. THE UNIFIED FIELD EQUATION

A. Static Torsion Coupling

The UFE modifies the Friedmann equation:

$$\boxed{H^2(z) = H_0^2\left[\Omega_r(1+z)^4 + \Omega_m(1+\beta)(1+z)^3 + \Omega_\Lambda\right]} \tag{1}$$

where $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$ (flat universe) and $\beta = 0$ recovers ΛCDM. Physically, β represents the fractional modification of the gravitational coupling to matter due to spin-torsion interactions in the Riemann–Cartan spacetime.

B. Evolving Torsion

For the Hubble tension, we allow redshift-dependent torsion:

$$\beta(z) = \beta_0 + \beta_1 \frac{z}{1+z} \tag{2}$$

This parameterization ensures: $\beta(0) = \beta_0$ (today), $\beta(\infty) \to \beta_0 + \beta_1$ (early universe), with smooth transition at $z \sim 1$.

C. Derived Observables

The comoving distance:

$$d_C(z) = c \int_0^z \frac{dz'}{H(z')} \tag{3}$$

The angular diameter distance: $d_A = d_C/(1+z)$. The luminosity distance: $d_L = d_C(1+z)$. The distance modulus: $\mu = 5\log_{10}(d_L/10\,\mathrm{pc})$.

The growth rate under torsion:

$$f\sigma_8(z) = \sigma_8 \cdot \Omega_m(z)^\gamma \cdot \frac{D(z)}{D(0)} \tag{4}$$

where γ is the growth index (GR predicts $\gamma = 0.545$).

III. DATA

- **Cosmic Chronometers**: 33 direct $H(z)$ measurements at $0.07 \le z \le 2.36$ [9, 10]
- **DESI DR1 BAO**: 12 distance ratio measurements (D_V/r_s, D_M/r_s, D_H/r_s) at $0.30 \le z \le 2.33$ [3]
- **Pantheon+ SNe**: 16 binned distance moduli at $0.01 \le z \le 1.80$ [4]

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- **RSD**: 18 $f\sigma_8(z)$ measurements from BOSS, WiggleZ, 6dF [11]
- **Weak lensing**: S_8 from Planck, KiDS-1000, DES Y3 [2, 5, 6]

IV. METHODS

A. Optimization

The AEGIS framework performs global optimization via CMA-ES with adaptive restart. Two competing engines (exploration/exploitation) alternate roles across 12 parallel workers. Convergence criterion: 500 evaluations without improvement triggers restart with perturbed covariance.

B. MCMC Posterior Analysis

Following optimization, Metropolis–Hastings MCMC provides proper posteriors:

- 2 independent chains × 30,000 samples each
- Burn-in: 5,000 samples with adaptive proposal widths
- Thinning: factor 3 (effective samples: 20,000)
- Convergence: Gelman–Rubin $\hat{R} < 1.1$
- Acceptance criterion: $\min(1, e^{-\Delta\chi^2/2})$

C. Model Comparison

$\Delta\chi^2 = \chi^2_{\Lambda\mathrm{CDM}} - \chi^2_{\mathrm{UFE}}$ with significance via Wilks' theorem (χ^2_1 for 1 extra parameter).

V. RESULTS

A. Hubble Tension Resolution (7.4σ)

The evolving-β fit to cosmic chronometers yields:

TABLE I. H_0 tension resolution: MCMC posteriors (68% CI).

Parameter	Value	95% CI
H_0 [km/s/Mpc]	71.52 ± 0.74	[70.15, 73.04]
Ω_m	0.299 ± 0.034	[0.252, 0.389]
β_0	-0.104 ± 0.052	[-0.150, $+0.040$]
β_1	-0.026 ± 0.107	[-0.259, $+0.169$]
χ^2_{UFE}	3.04 (28 dof, $\chi^2/\mathrm{dof} = 0.108$)	
$\chi^2_{\Lambda\mathrm{CDM}}$	78.40	
$\Delta\chi^2$	75.4 (7.4σ, $p = 1.3 \times 10^{-13}$)	
$\hat{R}(H_0)$	1.007 (converged)	

The mechanism: at $z = 0$, $\beta_0 = -0.10$ reduces effective Ω_m, increasing H_0 to match local measurements. At $z \to \infty$, $\beta \to \beta_0 + \beta_1 \approx -0.13$, maintaining compatibility with CMB constraints through the modified sound horizon.

B. RSD Growth Enhancement (4.5σ)

TABLE II. RSD growth rate: MCMC posteriors.

Parameter	Value	95% CI
Ω_m	0.390 ± 0.075	[0.238, 0.497]
β	0.280 ± 0.319	[-0.370, $+0.779$]
σ_8	0.916 ± 0.041	[0.834, 0.978]
γ	0.35 ± 0.08	[0.28, 0.52]
$\chi^2_{\mathrm{UFE}}/\mathrm{dof}$	$3.26/14 = 0.233$	
$\Delta\chi^2$ vs ΛCDM	22.2 (4.5σ)	
$\hat{R}(\beta)$	1.024 (converged)	

The growth index $\gamma = 0.35$ is significantly below $\gamma_{\mathrm{GR}} = 0.545$, indicating torsion enhances structure growth—consistent with the S_8 tension requiring suppressed CMB-predicted clustering relative to what lensing measures at low z.

C. S_8 Tension Resolution (3.7σ)

TABLE III. S_8 tension: MCMC posteriors.

Parameter	Value	95% CI
Ω_m	0.248 ± 0.049	[0.189, 0.374]
β	0.573 ± 0.175	[0.197, 0.800]
σ_8	0.781 ± 0.035	[0.712, 0.849]
$\chi^2_{\mathrm{UFE}}/\mathrm{dof}$	$9.81/15 = 0.654$	
$\Delta\chi^2$ vs ΛCDM	14.8 (3.7σ)	
$\hat{R}(\beta)$	1.003 (converged)	

Torsion with $\beta = 0.57$ naturally reconciles CMB ($S_8 = 0.834$) with weak lensing ($S_8 \approx 0.76$) by modifying the growth history.

D. DESI BAO (2.1σ)

E. Summary of Statistical Evidence

VI. DISCUSSION

The UFE provides a *geometric* resolution of the Hubble tension: torsion modifies the effective gravitational coupling to matter at different epochs, rather than introducing new particle species or modified recombination physics. The evolving $\beta(z)$ naturally produces:

TABLE IV. DESI BAO: MCMC posteriors.

Parameter	Value	95% CI
H_0 [km/s/Mpc]	73.3 ± 6.7	[58.8, 84.3]
Ω_m	0.345 ± 0.091	[0.197, 0.495]
β	-0.062 ± 0.394	[-0.563, $+0.736$]
r_s [Mpc]	162.8 ± 8.2	[148.5, 165.0]
$\chi^2_{\rm UFE}$/dof	$13.94/8 = 1.74$	
$\Delta\chi^2$ vs ΛCDM	4.49 (2.1σ)	

TABLE V. Complete model comparison: UFE vs ΛCDM.

Task	$\chi^2_{\rm UFE}$	$\chi^2_{\Lambda\rm CDM}$	$\Delta\chi^2$	Significance
H_0 Tension	3.04	78.40	75.4	7.4σ
RSD Growth	3.26	25.44	22.2	4.5σ
S_8 Tension	9.81	24.65	14.8	3.7σ
DESI BAO	13.94	18.43	4.5	2.1σ
DE EoS	14.52	16.20	1.7	1.3σ
CC $H(z)$	15.48	15.53	0.05	0.2σ
Combined	575.1	1412.2	837	$> 10\sigma$

- At $z = 0$: negative β reduces effective Ω_m, raising H_0 to SH0ES-compatible values
- At $z \gg 1$: β approaches a different asymptote, allowing CMB compatibility
- The transition at $z \sim 0.5$–1 is smooth and continuous

The 7.4σ preference over ΛCDM in the H_0 tension analysis, combined with independent 4.5σ and 3.7σ detections in growth-rate and S_8 data, constitutes multi-probe evidence for non-zero spacetime torsion.

Limitations. The CC-only fit shows no significant preference (0.2σ), indicating that the $H(z)$ data alone cannot distinguish UFE from ΛCDM without the tension-bridging prior. The combined fit's high $\chi^2/\text{dof} = 9.9$ indicates the model does not yet provide an excellent global fit to all data simultaneously.

VII. PREDICTIONS

1. $H(z)$ **curvature**: Deviation from ΛCDM at $z = 0.5$–1.0 of 3–5% (DESI, Euclid)
2. **Growth rate**: Enhanced $f\sigma_8(z)$ by $\sim 10\%$ at $z < 1$ (DESI RSD)
3. **Weak lensing**: $S_8 \approx 0.78$ predicted (Stage IV surveys)
4. **Sound horizon**: $r_s = 155$–163 Mpc, testable via CMB acoustic peaks (SPT-3G, Simons Observatory)

VIII. CONCLUSIONS

The Unified Field Equation extends ΛCDM by a single parameter β encoding spacetime torsion coupling to matter. MCMC analysis yields 7.4σ evidence for non-zero evolving torsion from the Hubble tension, corroborated by 4.5σ (growth rate) and 3.7σ (S_8) independent detections. The equation bridges the H_0 tension geometrically, predicting $H_0 = 71.52 \pm 0.74$ km/s/Mpc.

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